COMPUTER NETWORKS

MODULES 2, 3 & 4 – Remaining Syllabus Notes

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Study Notes for Examination Preparation

Prepared from the syllabus supplied by the student. This PDF contains Modules 2–4; Module 1 is available in a separate PDF.

Modules covered

- Module 2: Network Layer – routing, IP, QoS and Linux routing-table practice.
- Module 3: Data Link Layer – DLC, MAC, Ethernet, connecting devices, wireless LANs, Mobile IP and packet interfaces.
- Module 4: SNMP, ASN.1 and Physical Layer – data/signals, transmission and media.

MODULE 2 – NETWORK LAYER

1. Introduction to the Network Layer

The network layer is responsible for delivering packets from a source host to a destination host across one or more interconnected networks. It makes routing decisions and uses logical addressing, such as IP addresses.

Main functions

- Logical addressing – identifies source and destination hosts using IP addresses.
- Routing – selects a path through interconnected networks.
- Forwarding – moves a packet from an input interface to the appropriate output interface.
- Internetworking – enables communication across different networks.
- Fragmentation and reassembly – handles packet-size constraints where supported by the network protocol.

2. Network-Layer Service and Protocols

The network layer may offer connectionless or connection-oriented service models. The Internet Protocol (IP) provides connectionless, best-effort packet delivery: IP itself does not guarantee delivery, ordering or recovery.

Protocol / term	Purpose
IPv4	32-bit logical addressing and packet delivery.
IPv6	128-bit addressing and a redesigned packet format.
ICMP	Carries network control and error-reporting messages.
ARP (IPv4 networks)	Resolves a local IPv4 address to a link-layer address.
Routing protocol	Exchanges information used to calculate routes.

3. Unicast and Multicast Routing

Unicast routing

Unicast means one sender communicates with one destination. Routers examine the destination IP address and use their routing tables to forward packets along a selected path.

Multicast routing

Multicast means one sender sends data to a group of interested receivers. The network aims to distribute packets to group members without sending a separate full copy from the source to every receiver.

Unicast	Multicast
One sender to one destination	One sender to a group of receivers
Individual destination address	Multicast group address
Common in ordinary host-to-host communication	Useful for group streaming and conferencing

4. Intra-domain and Inter-domain Routing

Intra-domain routing operates within one autonomous system (AS), such as an organization's or service provider's network. Examples include RIP and OSPF.

Inter-domain routing operates between autonomous systems. BGP is the principal inter-domain routing protocol used on the Internet.

Intra-domain	Inter-domain
Within one AS	Between different ASes
Often focuses on internal path efficiency	Uses policy-based routing between networks
Examples: RIP, OSPF, IS-IS	Example: BGP

5. Routing Approaches

- Distance-vector routing: routers exchange distance information with neighbours and calculate routes using a distance metric. RIP is a familiar example.
- Link-state routing: routers distribute link-state information and independently calculate shortest paths. OSPF is a familiar example.
- Path-vector routing: advertises reachable network prefixes together with path and policy information. BGP uses this approach.

6. Next Generation IP – IPv6

IPv6 was developed to address limitations of IPv4, especially address-space exhaustion, and to support scalable Internet operation.

Feature	IPv4	IPv6
Address length	32 bits	128 bits
Address notation	Dotted decimal	Colon-separated hexadecimal
Header	Variable length (options)	Fixed basic header; extension headers
Broadcast	Supported	No broadcast; multicast is used
Address space	Smaller	Much larger

Example IPv4 address: 192.0.2.10. Example IPv6 address: 2001:db8::10. These are documentation/example addresses.

7. Quality of Service (QoS)

QoS refers to techniques used to provide predictable or differentiated network performance for selected traffic.

- Bandwidth – data-carrying capacity available to a flow.
- Delay – time taken for data to travel from source to destination.
- Jitter – variation in packet delay.
- Packet loss – packets that fail to reach the destination.

QoS mechanisms include traffic classification, scheduling, shaping, policing, admission control and resource reservation. Real-time voice and video are examples of applications sensitive to delay and jitter.

8. Linux Routing Table and ip Command – Practical

A routing table tells a Linux host or router where to send packets for different destination networks.

Useful commands

Command	Use
ip route	Display the IPv4 routing table.
ip -6 route	Display IPv6 routes.
ip route get 8.8.8.8	Show the route selected for a destination.
sudo ip route add 192.0.2.0/24 via 192.0.2.1	Add an example route through a gateway.
sudo ip route del 192.0.2.0/24	Delete the example route.

The addresses above are documentation examples. Use the correct network, gateway and interface values for the actual lab environment. Routes added with the ip command are commonly temporary and may not persist after reboot.

Module 2 – Important Questions

- Explain the functions of the network layer.
- Differentiate unicast and multicast routing.
- Explain intra-domain and inter-domain routing with examples.

- Compare distance-vector and link-state routing.
- Explain IPv6 and compare it with IPv4.
- Define QoS and explain its performance parameters.
- Explain the purpose of a Linux routing table and important ip route commands.

MODULE 3 – DATA LINK LAYER

1. Introduction to the Data Link Layer

The data link layer provides node-to-node delivery over a single link. It receives packets from the network layer, encapsulates them into frames and sends them over the physical medium.

Main functions

- Framing – divides a bit stream into identifiable frames.
- Physical addressing – uses link-layer addresses such as Ethernet MAC addresses.
- Error detection – detects transmission errors, commonly using a Frame Check Sequence (FCS).
- Flow control – may regulate the rate of transmission between directly connected nodes.
- Medium access control – coordinates access to a shared communication medium.

2. Data Link Control (DLC)

Data Link Control refers to mechanisms that support reliable or orderly communication over a link. Depending on the technology, these may include framing, flow control and error control.

Framing methods

- Character/byte count – a length field indicates the frame size.
- Byte stuffing – special bytes are inserted to distinguish data from frame delimiters.
- Bit stuffing – extra bits are inserted into the bit stream so that delimiter patterns are not mistaken for data.
- Physical-layer coding violations – special signal patterns may mark frame boundaries in some systems.

Error Detection

Error detection adds redundant information so that a receiver can identify corrupted frames.

- Parity check – adds a parity bit to make the number of 1s even or odd.
- Checksum – computes a value from data and checks it at the receiver.
- CRC (Cyclic Redundancy Check) – treats the bit sequence as a polynomial and calculates a remainder used for error detection.

3. MAC – Medium Access Control

MAC is the lower sublayer of the data link layer in the IEEE 802 model. It controls how devices share and access a common transmission medium and handles MAC addressing.

- MAC address – link-layer identifier, commonly 48 bits in Ethernet.
- Unicast MAC – identifies one interface.
- Multicast MAC – identifies a group of interfaces.
- Broadcast MAC in Ethernet – FF:FF:FF:FF:FF:FF.

4. Link Addressing

A link-layer address is used to deliver a frame on the local network segment. It is different from an IP address: IP addresses support logical, network-wide addressing, while MAC addresses are used for local link delivery.

IP address	MAC address
Logical network-layer address	Link-layer address
Used for routing between networks	Used for delivery on a local link
IPv4: 32 bits; IPv6: 128 bits	Ethernet MAC: commonly 48 bits

5. Ethernet

Ethernet is a widely used family of wired LAN technologies standardized mainly by IEEE 802.3.

Ethernet frame – simplified fields

Field	Purpose
Preamble and SFD	Help receiver synchronize and identify frame start.
Destination MAC	Address of intended receiving interface/group.
Source MAC	Address of sending interface.
Type/Length	Identifies upper-layer protocol or payload length, depending on frame format.
Payload and padding	Carries data; padding may ensure minimum frame size.
FCS	CRC-based error-detection field.

Traditional shared Ethernet used CSMA/CD. Modern switched full-duplex Ethernet generally does not use collision detection because each link is a separate full-duplex connection.

6. Connecting Devices

Device	Main function
Repeater	Regenerates signals to extend a physical link; operates at the physical layer.
Hub	Multiport repeater; sends incoming bits to other ports.
Bridge	Connects LAN segments and filters frames using MAC addresses.
Switch	Multiport bridge that forwards frames based on a MAC address table.
Router	Connects IP networks and forwards packets using network-layer information.
Gateway	General term for a device/system that connects dissimilar networks or performs protocol conversion.

7. Wireless LANs

A Wireless Local Area Network (WLAN) connects devices using radio communication instead of physical Ethernet cables. Wi-Fi is based on the IEEE 802.11 family.

Basic components

- Station (STA) – a device with a wireless network interface.
- Access Point (AP) – connects wireless stations to a distribution system or wired network.
- Basic Service Set (BSS) – a group of stations associated with an AP, or an independent group in ad hoc mode.
- SSID – the network name used to identify a WLAN.

Wireless access and security

Wi-Fi commonly uses carrier sensing and collision avoidance (CSMA/CA), rather than Ethernet's traditional CSMA/CD. Authentication and encryption help protect WLAN access; WPA2 and WPA3 are widely used security standards.

8. Mobile IP

Mobile IP is a set of mechanisms designed to allow a mobile node to move between networks while maintaining reachability through a stable home address.

- Mobile node – device that changes its point of attachment to the network.
- Home network – network associated with the mobile node's home address.
- Home agent – entity in the home network that helps maintain reachability.
- Care-of address – address representing the mobile node's current location in a visited network.
- Tunnelling – may be used to forward packets from the home network toward the mobile node's current location.

Conceptual operation: the mobile node moves to a visited network, obtains a care-of address, registers its location as required, and packets are forwarded/tunnelled toward the current location.

9. Data Link Provider Interface (DLPI)

DLPI is a UNIX System V style interface that allows a user-level process to access data-link services provided by a data-link provider. It defines a standard interface between data-link users and providers.

- Supports access to data-link services through defined primitives.
- Helps applications bind to a data-link service access point and exchange link-layer data.
- Is relevant to systems and networking environments that implement the DLPI interface.

10. SOCK_PACKET and PF_PACKET

These are Linux packet-socket mechanisms used by programs that need access to link-layer packets.

Interface	Description
SOCK_PACKET	Older Linux packet-socket interface; considered obsolete on modern systems.
PF_PACKET	Linux protocol family for packet sockets, allowing applications to receive or transmit packets at the device/ link layer.

PF_PACKET is commonly used by packet analyzers and low-level networking tools. Access may require elevated privileges. Modern Linux applications generally prefer PF_PACKET over the legacy SOCK_PACKET interface.

Module 3 – Important Questions

- Explain the functions of the data link layer.
- Describe framing methods and error-detection techniques.
- Explain MAC addressing and distinguish MAC address from IP address.
- Draw and explain the Ethernet frame format.
- Compare repeater, hub, bridge, switch and router.
- Explain the architecture and operation of a wireless LAN.
- Explain Mobile IP and the roles of home agent and care-of address.
- Write short notes on DLPI, SOCK_PACKET and PF_PACKET.

MODULE 4 – SNMP, ASN.1 AND PHYSICAL LAYER

1. SNMP – Simple Network Management Protocol

SNMP is an application-layer protocol used to monitor and manage network devices such as routers, switches, servers and printers.

SNMP architecture

- Managed device – network device being monitored or managed.
- Agent – software running on or associated with the managed device; gathers management information.
- Network Management Station (NMS) – management system that communicates with agents.
- MIB (Management Information Base) – structured collection of managed objects and their identifiers.

SNMP operations

Operation	Purpose
GET	Manager requests the value of an object.
GET-NEXT	Requests the next object in MIB order; used in table traversal.
GET-BULK	Retrieves larger sets of management information efficiently (SNMPv2 and later).
SET	Manager requests a change to a managed object.
RESPONSE	Agent returns a result or requested information.
TRAP	Agent sends an unsolicited event notification to the manager.
INFORM	Acknowledged notification mechanism.

SNMP commonly uses UDP: port 161 for manager-agent requests and responses, and port 162 for notifications such as traps. SNMPv3 adds important security features including authentication and privacy.

2. ASN.1 – Abstract Syntax Notation One

ASN.1 is a standard notation used to define data structures in a machine-independent way. It describes the abstract structure of information so that systems from different vendors can exchange data.

Encoding rules

- BER (Basic Encoding Rules) – flexible encoding rules using tag, length and value fields.
- DER (Distinguished Encoding Rules) – a restricted form of BER that produces a unique encoding for a given value.
- PER (Packed Encoding Rules) – designed for compact encodings.

SNMP uses a subset of ASN.1 to define managed objects and uses BER encoding in traditional SNMP message formats.

3. Physical Layer – Introduction

The physical layer is the lowest layer of the OSI model. It transmits raw bits over a communication medium and defines the electrical, optical, radio and mechanical characteristics of the interface.

Functions

- Defines physical interfaces, connectors and transmission media.
- Specifies signal representation of bits.
- Defines data rate and bit timing.
- Specifies transmission mode and physical topology.
- Defines how devices are connected to the medium.

4. Data and Signals

Data is the information to be communicated. A signal is the physical representation of data that travels through a transmission medium.

Term	Meaning
Analog data	Information that varies continuously, such as voice.
Digital data	Information represented using discrete values, commonly binary 0 and 1.
Analog signal	Continuous waveform that varies with time.
Digital signal	Signal represented by discrete levels or pulses.

5. Digital and Analog Transmission

Digital transmission

Digital transmission sends information using discrete signal elements or pulses. Repeaters can regenerate digital signals, helping reduce the accumulation of noise over long distances.

Analog transmission

Analog transmission uses continuously varying signals. Amplifiers may strengthen the signal, but they can also amplify noise along with it.

Digital transmission	Analog transmission
Uses discrete signal levels	Uses continuous waveforms
Can be regenerated by repeaters	Often strengthened using amplifiers
Common in digital communication systems	Common in traditional radio and analog systems

6. Signal Characteristics

- Amplitude – strength or height of a signal.
- Frequency – number of cycles per second, measured in hertz (Hz).
- Period – time required for one complete cycle; T = 1/f.
- Phase – position of a waveform relative to a reference.
- Wavelength – physical distance covered by one cycle; for a wave, wavelength = propagation speed / frequency.

For a periodic signal, frequency and period are reciprocals: f = 1/T.

7. Bandwidth Utilization

Bandwidth utilization techniques aim to use the available channel capacity efficiently.

Multiplexing

- FDM (Frequency Division Multiplexing) – divides channel bandwidth into frequency bands for different signals.
- TDM (Time Division Multiplexing) – assigns different time slots to different data streams.
- WDM (Wavelength Division Multiplexing) – carries multiple optical signals at different wavelengths through a fibre.

Spread Spectrum

Spread-spectrum techniques spread a signal over a bandwidth wider than the minimum required. They can improve resistance to interference and support multiple users or secure communication designs.

- FHSS (Frequency Hopping Spread Spectrum) – changes carrier frequency according to a hopping sequence.
- DSSS (Direct Sequence Spread Spectrum) – combines data with a higher-rate spreading code.

8. Transmission Media

Transmission media are the physical paths through which signals travel. They are broadly classified as guided and unguided media.

Guided media

Medium	Features
Twisted-pair cable	Two insulated copper wires twisted together; inexpensive and widely used in LANs.
Coaxial cable	Central conductor with insulation and shielding; used in cable networks and other systems.
Optical fibre	Carries light through glass/plastic fibre; high bandwidth, low attenuation and immunity to electromagnetic interference.

Unguided media

Medium	Features
Radio waves	Can travel over broad areas; used in broadcasting and wireless networks.
Microwaves	Directional, often line-of-sight; used in point-to-point links and satellite communication.
Infrared	Short-range communication; generally does not pass through walls.
Satellite links	Use satellites to relay signals over large geographic areas.

Comparison of guided media

Feature	Twisted pair	Coaxial	Optical fibre
Signal	Electrical	Electrical	Light
EMI effect	More susceptible	Shielding improves resistance	Immune to electromagnetic interference
Bandwidth	Moderate to high, depends on category	Moderate to high, depends on type	Very high
Cost/installation	Usually economical and easy	Moderate	Equipment and termination can cost more

Module 4 – Important Questions

- Explain SNMP architecture and its components.
- Describe the major SNMP operations.
- What is ASN.1? Explain BER and DER.
- Explain the functions of the physical layer.
- Differentiate analog data, digital data, analog signals and digital signals.
- Compare digital and analog transmission.
- Explain FDM, TDM and WDM.
- Differentiate guided and unguided transmission media.
- Explain twisted-pair, coaxial and optical-fibre cables.
- Write short notes on radio waves, microwaves and satellite communication.

Final Revision – Modules 2 to 4

Module	Must-revise areas
2 – Network Layer	Routing functions; unicast/multicast; intra/inter-domain; IPv4 vs IPv6; QoS; Linux ip route.
3 – Data Link Layer	Framing; error detection; MAC; Ethernet frame; network devices; WLAN; Mobile IP; DLPI/PF_PACKET.
4 – SNMP & Physical	SNMP components/operations; ASN.1/BER; signal characteristics; multiplexing; guided/unguided media.

Study guidance: Practise labelled diagrams for OSI layering, Ethernet frame, SNMP architecture and transmission media. For comparison questions, prepare tabular differences. For Linux practical questions, understand what each command does and use only valid network values supplied by your lab instructor.

Disclaimer: This is a student study aid based on the syllabus supplied in the conversation. It is not an official KTU publication. The syllabus mentions selected practical interfaces and commands; exact implementation details can vary by operating system and lab environment. Confirm detailed procedures with your prescribed textbook and instructor.